

SHAFAYET KHAN SHAFEE

✉ sshafee@isrt.ac.bd 🌐 kshafayet.netlify.app 📄 [shafayetShafee](#) [in shafayetshafee](#) 🎓 [Google Scholar](#)
☎ +880-1791-051104 📍 Dhaka, Bangladesh

Statistician and data scientist with applications in public health and industry. Committed to methodological rigor, reproducible research, and open-source software development.

RESEARCH INTERESTS

Causal Inference, Multilevel Modeling, Bayesian Inference, Survival Analysis, Sensitivity Analysis, Statistical Machine Learning.

EDUCATION

M.Sc. in Applied Statistics Dhaka, Bangladesh
GPA 3.97 / 4.00 2022–2023
Institute of Statistical Research & Training
University of Dhaka

- **Thesis:** Extended the median odds ratio framework for quantifying between-cluster variation in hierarchical binary data with more than two levels and developed corresponding interval estimators.
- **Selected Coursework:** Causal Inference; Multilevel Modeling; Bayesian Inference; Spatial Statistics; Statistical Machine Learning.

B.Sc. in Applied Statistics Dhaka, Bangladesh
CGPA 3.96 / 4.00 2018–2022
Institute of Statistical Research & Training
University of Dhaka

- **Project:** Examined the association between caregivers' stimulation activities and socio-emotional development in under-five children using nationally representative MICS 2019 Bangladesh data and logistic regression analysis.
- **Selected Coursework:** Statistical Inference; Multivariate Statistics; Generalized Linear Models; Analysis of Time Series; Design of Experiments; Epidemiology; Econometrics.

WORK EXPERIENCE

Data Scientist Dhaka, Bangladesh
Pathao Pay Jan 2025–Present

- Applied Bayesian experimental design (Bayes Factor Design Analysis) to estimate sample sizes for controlled product rollouts, incorporating prior information and evidence-based stopping criteria.
- Estimated causal effects of a fintech product rollout on merchant transaction behavior using propensity score matching and Difference-in-Differences, with explicit attention to covariate balance diagnostics.
- Used Hierarchical Bayesian modeling to evaluate adoption rates across cohorts in a controlled rollout, accounting for small and unequal group sizes and propagating uncertainty through to product decisions.
- Built scalable analytical pipelines using dbt, BigQuery, and cloud infrastructure to support reproducible statistical analysis and reporting.

Data Scientist Dhaka, Bangladesh
Pathao Limited Jul 2023–Dec 2024

- Applied causal ML methods—Double ML and generalized random forests—to estimate heterogeneous treatment effects across user subgroups, connecting industry experimentation practice to the causal inference literature.

- Co-developed a repayment risk model using Random Survival Forests to estimate time-to-repayment distributions, drawing on survival analysis methods for a credit risk application.
- Designed and analyzed randomized controlled experiments to evaluate product initiatives, including cohort construction and randomization checks.

PUBLICATIONS

† Corresponding author * Equal contribution

- [1] S. K. Shafee[†] and M. S. Rahman. *On the estimation of the median odds ratio for measuring contextual effects in multilevel binary data from complex survey designs*. [Under review]. June 2026. arXiv: 2606.15145 [stat.ME]. URL: <https://arxiv.org/abs/2606.15145>.
- [2] S. K. Shafee[†], B. Sarker*, and M. N. I. Sium*. *G-computation for causal effect estimation from observational hierarchical data with unmeasured cluster context*. [Under review]. June 2026. arXiv: 2606.14131 [stat.ME]. URL: <https://arxiv.org/abs/2606.14131>.
- [3] S. K. Shafee[†], M. N. I. Sium, B. Sarker, and R. Islam. “Investigating the causal effect of maternal continuum of care on child’s minimum acceptable diet: A multilevel approach using partially pooled propensity score weighting”. In: *PLOS ONE* 20.11 (Nov. 2025), pp. 1–13. DOI: 10.1371/journal.pone.0335972.

HONORS & AWARDS

- **Dean’s Award**, *Faculty of Science, University of Dhaka* 2025
- **Conference Award for Scientists**, for the abstract “Interval Estimation of the Median Odds Ratio for Measuring Contextual Effects in Multilevel Data Using a Binary Logistic Model”, *45th Annual Conference of the International Society for Clinical Biostatistics (ISCB45)*, Thessaloniki, Greece. 2024
- **National Science and Technology (NST) Fellowship**, for M.Sc. thesis research on multilevel modeling, *Ministry of Science and Technology (MoST), Government of Bangladesh* 2023

TALKS & PRESENTATIONS

- On the Estimation of the Median Odds Ratio for Measuring Contextual Effects in Multilevel Binary Data from Complex Survey Designs, *International Conference on Applied Statistics and Data Science (ICASDS 2025)*, University of Dhaka 2025

TECHNICAL SKILLS

- **Machine Learning**: Ensemble Models, Causal ML (DML/GRF), Model Calibration, Conformal Prediction
- **Data & MLOps**: BigQuery, dbt, Kedro, MLflow, Docker, Bash.
- **Analytics Tools**: Stata, SPSS, Shiny, Looker Studio, Mixpanel.
- **Programming**: R, Python, Octave, SQL, Lua.
- **Others**: L^AT_EX, Git, GitHub Actions, GitLab CI.

SOFTWARE

- **relativity-simulator**, *Personal project* 1920–1930
- **quantum-probability**, *Collaborative research* 1924–1935